CS39006: Networks Laboratory
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Assignment 7: Lightweight Custom Discovery Protocol (CLDP) Specification

Name: Praveen Kumar
Roll Number: 22CS10054

1. Introduction

The Lightweight Custom Discovery Protocol (CLDP) is a custom protocol designed for a closed network environment to enable nodes to announce their presence and query metadata from other nodes. This specification outlines the packet format, message types, and design details implemented using raw sockets in POSIX C.

2. Protocol Overview

CLDP operates at the IP level with a custom protocol number (253). It does not rely on transport-layer protocols like TCP or UDP. The protocol supports three message types:
- 0x01: HELLO (announces a node's presence)
- 0x02: QUERY (requests metadata from other nodes)
- 0x03: RESPONSE (responds with requested metadata)

3. Packet Format

Each CLDP packet consists of an IP header followed by a custom CLDP header and an optional payload.

3.1 IP Header

- Version: 4 (IPv4)
- IHL: 5 (20 bytes, no options)
- Total Length: Variable (depends on payload size)
- Protocol: 253 (custom protocol number)
- Source/Destination IP: Set by sender (e.g., 172.30.233.130 or broadcast 255.255.255.255)
- Other fields (TTL, ID, etc.) are set as per standard IP conventions.

3.2 CLDP Header (8 bytes)

Field	size(bytes)	Description
Message Type	1	0x01 (HELLO), 0x02 (QUERY), 0x03 (RESPONSE)
Payload Length	1	Length of payload in bytes (0 if none)
Transaction ID	2	Unique identifier for request-response pairing
Reserved	4	Set to 0 (for future use)

3.3 Payload (Variable Length)

- HELLO: Contains the hostname string (e.g., "perfecta").
- QUERY: Empty (metadata request is implicit: hostname and timestamp).
- RESPONSE: Contains hostname (variable-length string) followed by timestamp (struct timeval, 16 bytes).

Example Packet (HELLO):
- IP Header (20 bytes): [version=4, ihl=5, protocol=253, ...]
- CLDP Header (8 bytes): [msg_type=0x01, payload_len=7, transaction_id=1, reserved=0]
- Payload (7 bytes): "perfecta"

4. Supported Metadata

The implementation supports the following metadata in RESPONSE messages:
1. Hostname: Retrieved via gethostname() (e.g., "perfecta").
2. Timestamp: Retrieved via gettimeofday() (struct timeval with seconds and microseconds).
3. (Optional third metadata could be added, e.g., CPU load via sysinfo(), but only hostname and timestamp are implemented as per demo output.)

5. Design Details

- Raw Sockets: Implemented using AF_INET and SOCK_RAW with protocol number 253.
- IP Header Crafting: Enabled via setsockopt() with IP_HDRINCL.
- Broadcasting: QUERY and HELLO messages are sent to 255.255.255.255 for network-wide discovery.
- Packet Parsing: Server filters packets by protocol number (253) and processes only CLDP messages.
- Error Handling: Basic checks for socket creation and send/receive operations.

6. Demonstration Scenario

- Servers announce themselves with HELLO every 10 seconds.
- Clients send QUERY messages to request hostname and timestamp.
- Servers respond with RESPONSE packets containing the requested metadata.